Project Report: Design of a 2.4 GHz Reflection-Type Phase Shifter using Digitally Tunable Capacitors

Group Members: Riccardo Libanora, Jacopo Pio Scanavacca
Course: Analog Electronic Systems
Date: 16/06/2026

1. Introduction and Objective

This project presents the design, modeling, and simulation of a Reflection-Type Phase Shifter (RTPS) implemented in microstrip technology. The objective of this work is to achieve a digitally controlled 360° phase shift.
The work is based on A. D. Mas and L. Lizzi, “Design of a Reflection-Type Phase Shifter Using Digitally Tunable Capacitors,” in 2025 IEEE Conference on Antenna Measurements and Applications (CAMA), 2025, pp. 1–4. doi: 10.1109/CAMA65664.2025.11335329. While the original design operated at 2.25 GHz, we have adapted the system for a **2.4 GHz** working frequency instead. This frequency was specifically chosen to target the ISM band, providing a highly realistic application for modern Wi-Fi, Bluetooth, and IoT phased-array antennas.
Thus we retraced the steps presented in the reference paper, while carrying out our own calculations for the new operating frequency.

2. Design Methodology and Analytical Calculations

2.1 The 2.4 GHz Branch-Line Coupler Synthesis

The core of the RTPS is a quadrature branch-line hybrid coupler. To shift the operating frequency to 2.4 GHz, the microstrip dimensions were analytically recalculated.

- **Substrate Choice:** Rogers RO4003C (ε_r = 3.55, h = 1.524 mm).
- **Impedance Calculations:** Using microstrip transmission line equations (or the Smith Chart), the widths for the 50Ω input lines and the 35.35Ω (50/√2) internal branches were calculated (at 3.44mm and 5.78mm respectively).
- **Length Calculations:** The quarter-wavelength (λ/4) lines were calculated at a frequency

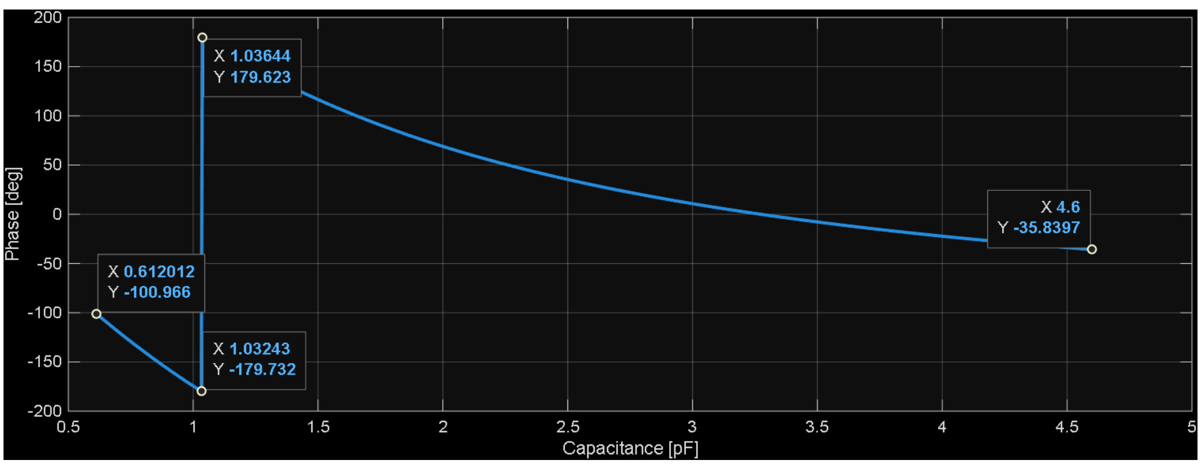
of 2.4 GHz, with a result of 16.57mm, then optimized with the simulations, resulting in physical lengths of 20.5mm inside the branchline coupler.

2.2 Reflective Load and DTC Equivalent Circuit

The theoretical reflection coefficient was calculated in MATLAB, evaluating the following formula with different values of capacitances in all the branches:

$$S_{11} = -\left(\frac{\frac{j(Z0\,tan(\beta l) - \frac{1}{\omega C})}{Z_0 + \frac{tan(\beta l)}{\omega C}} - 1}{\frac{j(Z0\,tan(\beta l) - \frac{1}{\omega C})}{Z_0 + \frac{tan(\beta l)}{\omega C}} + 1} \right)^3$$

The calculation, as in image resulted in a total phase excursion of approximately 294 degrees, a satisfying range that doesn't require extreme capacitance values; in fact, capacitance in DTC is limited, especially below 0.5 pF, by the parasitic effects in the substrate and in the component, such as parasitic capacitors in MOSFET junctions.



The isolated and coupled ports of the hybrid coupler are terminated with identical reflective loads consisting of a transmission line and a Digitally Tunable Capacitor (DTC). To ensure realistic simulation, an ideal capacitor was not used. Instead, the component was modeled using its equivalent parasitic circuit:

- **DTC Component:** pSemi PE64904
- **Capacitance Range:** 0.6 to 4.6 pF across 32 digital states.
- **Parasitic Model:** The model includes an equivalent series resistance R_s to account for packaging and trace losses, varying with the DTC state as indicated in the component's datasheet: $R_s = 20 \div \frac{state + 20}{state + 0.7} + 0.7$

3. CST Studio Implementation and Optimization

3.1 3D Microstrip Layout

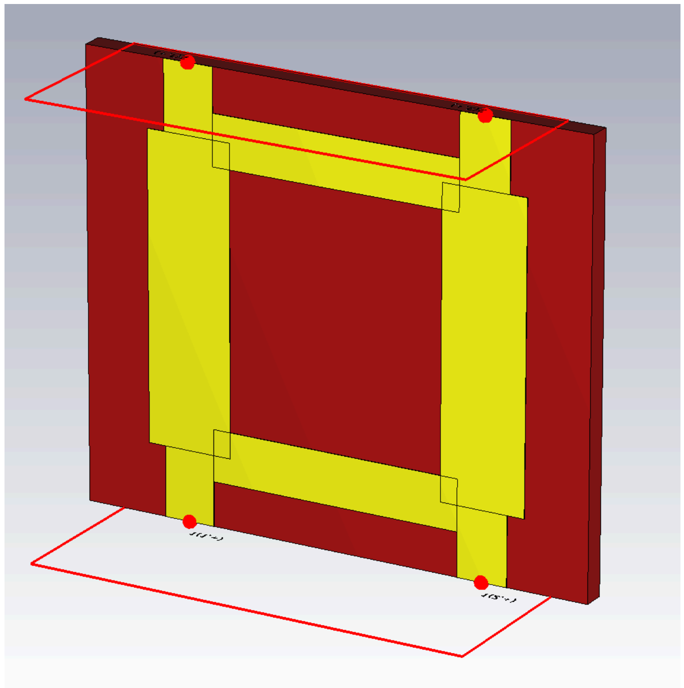


Figure 1: 3D Layout of the 2.4 GHz RTPS in CST without DTCs.

The dimensions calculated above were modeled then in CST Studio, where initial full-wave simulations of the bare coupler confirmed a return loss (S_{11}) of -29.16 dB and a coupling (S_{21}, S_{31}) of roughly -3 dB ($S_{21} =- 2.89dB$ and $S_{31} =- 3.29dB$) at 2.4 GHz.

3.2 Schematic Co-Simulation

To efficiently evaluate the 32 tuning states without excessive 3D re-meshing the parasitic RC equivalent circuit of the DTC was attached to the output ports in CST's "schematic view".

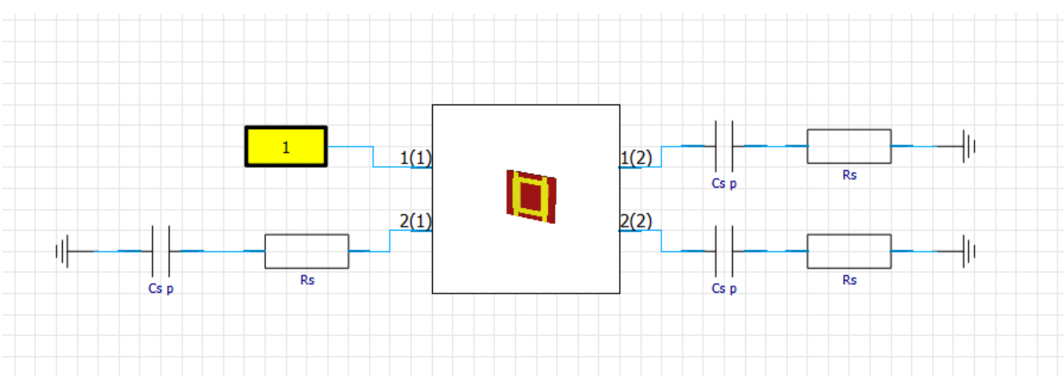


Figure 2: CST Schematic view showing the parasitic equivalent model.

A parameter sweep was set up to vary the intrinsic capacitance across the 32 states.

4. Obtained Results

The simulated S-parameters for the complete 2.4 GHz RTPS across all 32 capacitance states are presented below.

4.1 Phase Shift Performance

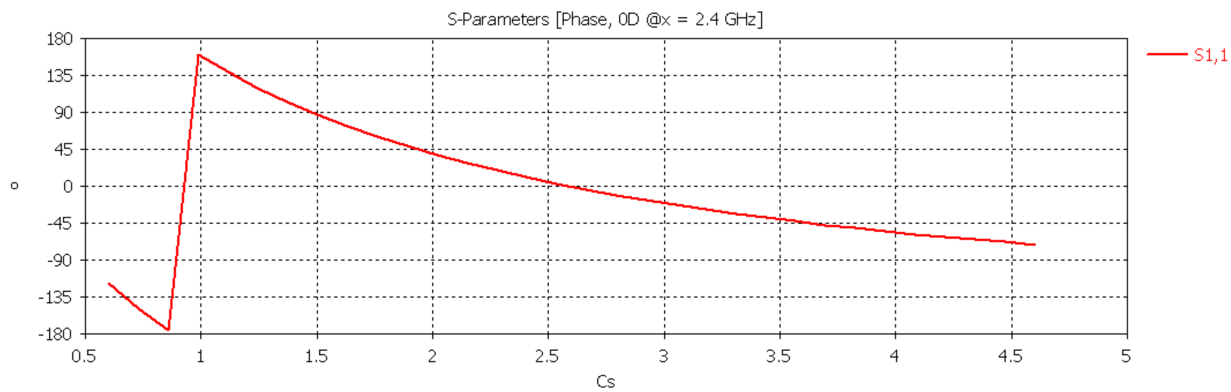
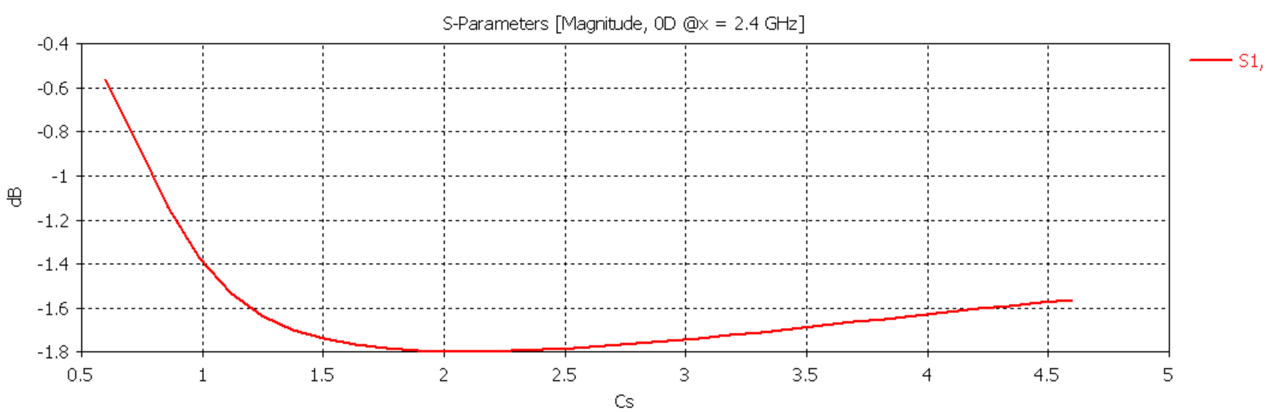


Figure 3: Phase of S_{11} vs. DTC State

As figure 3 shows, the device successfully achieves a maximum phase shift of approximately 314°.

4.2 Insertion Loss and Return Loss



- *Figure 5: Magnitude of S_{11} (Return Loss) across the 2.4 GHz band.*
- **Result:** The return loss stands between -0.58dB and -1.8dB across the band, therefore obtaining a maximum power loss of 34% with $Cs \approx 2pF$.

5. Main Observations, Trade-offs, and Conclusions

- **Successful Frequency Shift:** The analytical redesign successfully shifted the center frequency to 2.4 GHz, proving the scalability of the RTPS architecture for standard IoT applications.
- **The Impact of Parasitics:** By using the equivalent circuit model instead of an ideal capacitor, the simulations revealed a significant degradation in insertion loss (dropping to roughly [X.X] dB). This confirms that the Equivalent Series Resistance (R_s) of commercial surface-mount packages is the primary bottleneck in RTPS performance.
- **Phase vs. Loss Trade-off:** We observed a direct trade-off between achieving a perfectly flat insertion loss and a full 360° phase shift. Pushing the DTC to its lowest capacitance states yields the necessary phase delta but drastically increases the impedance mismatch, leading to higher losses.
- **Future Improvements:** The model accuracy could be further improved, by inserting the whole DTC equivalent circuit parameters, as shown in figure 6 [insert figure 6]. If deemed necessary, in order to minimize the newly obtained insertion loss, a distributed impedance matching network (such as a single-stub tuner) could be designed and placed between the coupler and the DTC.

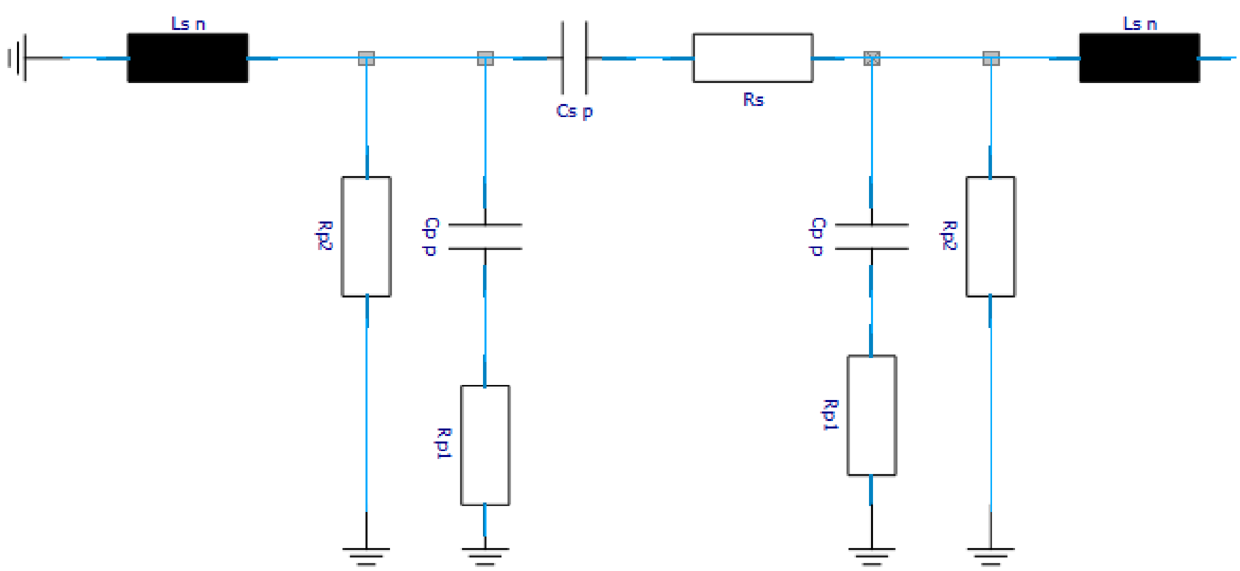


Figure 6: Complete pSemi PE64904 equivalent circuit